AP CALCULUS AB

Unit 1 • Lesson 1.8 practice

Lesson 1.8 • Determining Limits Using the Squeeze Theorem

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 0} x \sin \frac{1}{x}$ is

(A) 0

(B) 1

(C) It does not exist.

(D) ∞

Question 2

NO CALCULATOR

$\lim_{x \to 0} \frac{\sin 4x}{2x}$ is

(A) 4

(B) 2

(C) $\frac{1}{2}$

(D) 1

Question 3

NO CALCULATOR

$\lim_{x \to 0} \frac{1 - \cos x}{x}$ is

(A) 1

(B) -1

(C) 0

(D) It does not exist.

Question 4

NO CALCULATOR

If $3 - x^2 \leq f(x) \leq 3 + x^2$ for all x, then $\lim_{x \to 0} f(x)$ is

(A) 0

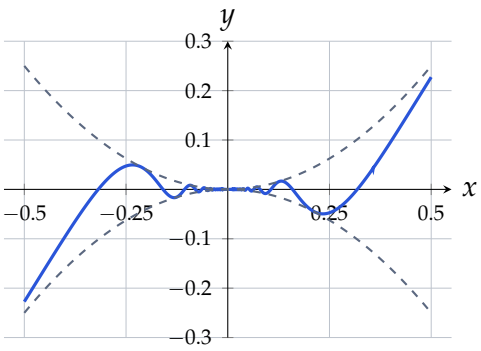
(B) 6

(C) It cannot be determined.

(D) 3

Question 5 (graph) NO CALCULATOR

The graph of $f(x) = x^2 \sin \frac{1}{x}$ is shown between the dashed graphs of $y = x^2$ and $y = -x^2$. What is $\lim_{x \to 0} f(x)$?



(A) 0
(B) 1
(C) It does not exist because f oscillates.
(D) -1

Question 6 (Free response) NO CALCULATOR

(a) Explain why $-|x| \leq x \cos \frac{1}{x} \leq |x|$ for $x \neq 0$, and use it to find $\lim_{x \to 0} x \cos \frac{1}{x}$.
(b) Find $\lim_{x \to 0} \frac{\tan x}{x}$.
(c) Find $\lim_{x \to 0} \frac{x}{\sin 3x}$.